

3-Tier architecture

To create the 3 Tier architecture requirements:

- 1) Create 3 Virtual Machines.
 - VM-WEB
 - VM-APP
 - VM-DB
- 2) TOMCAT installation process.
- 3) MySQL DB installation process.

1) Created 3 Virtual Machines:

- WEB VM
- APP VM
- DB VM

Name	Subscription	Resource group	Location	Status	Operating system	Size	Public IP address	Disks	Private
APP-VM	Free Trial	MRG-O1	Canada Central	Running	Linux	Standard_B1s	4.174.218.91	1	10.0.2.4
DB-VM	Free Trial	MRG-O1	Canada Central	Running	Linux	Standard_B1s	4.206.171.139	1	10.0.3.4
WEB-VM	Free Trial	MRG-O1	Canada Central	Running	Linux	Standard_B1s	4.205.168.240	1	10.0.1.4

In WEBSERVER install nginx:

Connect to WEB-VM through Putty using public IP address

1. sudo su
2. apt update
3. apt install nginx -y

while creating VM we can install nginx

put this in advance settings

Custom data	<pre>#!/bin/bash sudo su apt update apt install nginx -y</pre>
-------------	--

2)Tomcat installation process.

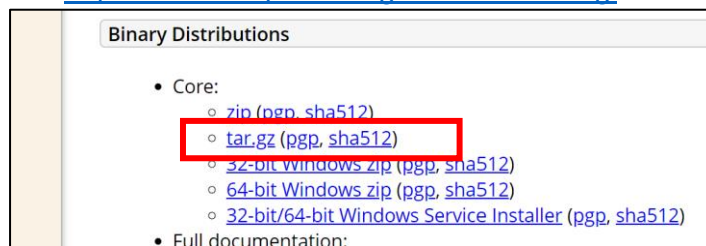
TOMCAT INSTALLATION ON UBUNTU:

Connect to APP-VM through Putty tool using public ip address.

1. sudo su
2. apt update -y
3. apt install maven
4. apt install openjdk-17-jdk -y

Go to google and browse tomcat 10 download or follow through the below link:

<https://tomcat.apache.org/download-10.cgi>



copy the path from ◦ [tar.gz \(pgp, sha512\)](#)

5. wget <https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.28/bin/apache-tomcat-10.1.28.tar.gz> (past the url)
6. ls
7. tar -xvzf apache-tomcat-10.1.28.tar.gz
8. ls
9. mv apache-tomcat-10.1.28 tomcat
10. ls
11. cd tomcat/
12. ls
13. cd webapps/
14. ls
15. cd host-manager/
16. ls
17. cd META-INF/
18. ls
19. vi context.xml

<!-- Valve className="org.apache.catalina.valves.RemoteAddrValve"

allow="127.\d+.\d+.\d+|::1|0:0:0:0:0:0:1" / -->

enter !-- & -- (as highlighted above)

20. cd ..
21. cd ..
22. cd ..
23. cd conf/
24. ls
25. vi tomcat-user.xml (enter tom&tab to fill)

copy below code & past as shown below fig1:

code

```
<role rolename="manager-gui" />
```

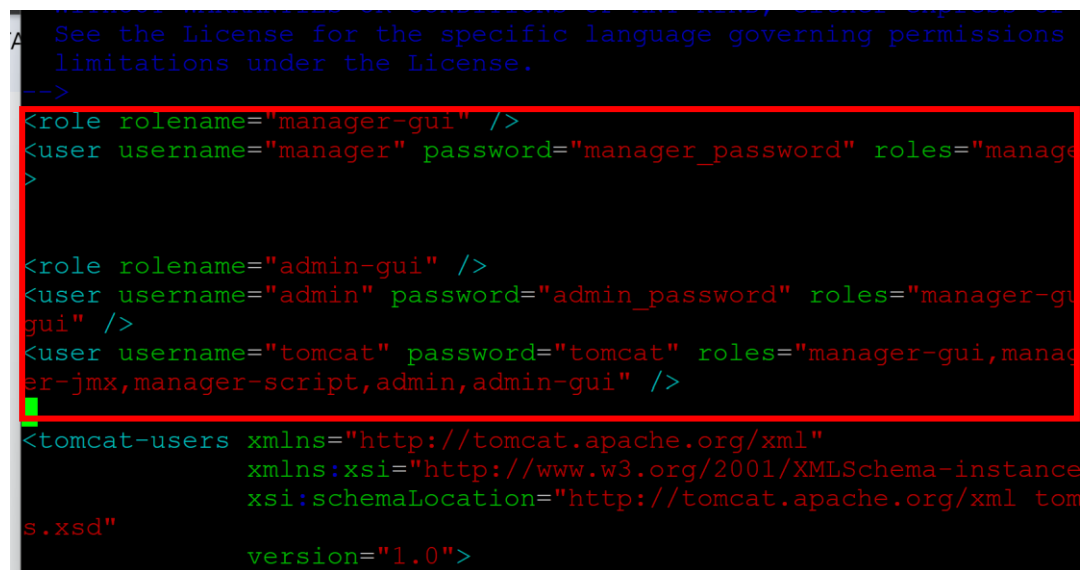
```
<user username="manager" password="manager_password" roles="manager-gui" />
```

```
<role rolename="admin-gui" />
```

```
<user username="admin" password="admin_password" roles="manager-gui,admin-gui" />
```

```
<user username="tomcat" password="tomcat" roles="manager-gui,manager,manager-jmx,manager-script,admin,admin-gui" />
```

FIG1:



```
-->
See the License for the specific language governing permissions
limitations under the License.
-->
<role rolename="manager-gui" />
<user username="manager" password="manager_password" roles="manager-gui" />

<role rolename="admin-gui" />
<user username="admin" password="admin_password" roles="manager-gui,admin-gui" />
<user username="tomcat" password="tomcat" roles="manager-gui,manager-jmx,manager-script,admin,admin-gui" />

<tomcat-users xmlns="http://tomcat.apache.org/xml"
               xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
               xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
               version="1.0">
```

26. cd ..
27. ls
28. cd webapps/
29. ls
30. cd manager/
31. ls
32. cd META-INF/
33. ls
34. vi context.xml

```
<!-- Valve className="org.apache.catalina.valves.RemoteAddrValve"
```

```
allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:1" / -->
```

➤ enter !-- & -- (as highlighted above)

35. cd ..
36. cd ..
37. cd ..
38. cd conf/
39. ls
40. cd ..

41. ls
42. cd bin
43. ls
44. ./shutdown.sh
45. ./startup.sh
46. Cd ../../

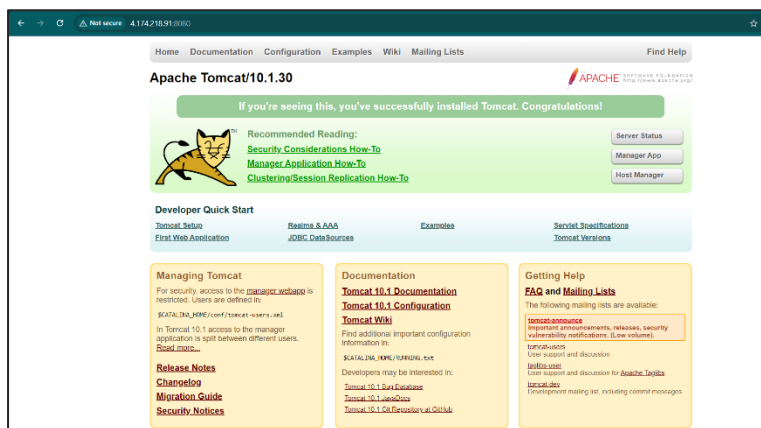
Installations completed

Enter public IP address with : & 8080(ex:-172.167.208.125)

Check it was working or not.

Installations completed successful

Connected to APP-VM with public IP:



4) MySQL DB Installation Process.

install MySQL DB on Linux machine using the below commands:

- Sudo su – root user.
- 1. apt update —update the machine.
- 2. apt install mysql-server -y ---- install my sql server.
- 3. systemctl start mysql.service --- start the my sql server.
- 4. systemctl status mysql ---- check the status whether it is in active & running or not.
- 5. systemctl enable mysql ----- enable my sql in your system.
- 6. mysql_secure_installation .
- 7. y
- 8. y
- 9. 2

10. Y
11. Y
12. y
13. mysql --- login into the mysql & check the , exit from my sql.
14. ctrl+z.
15. nano /etc/mysql/mysql.conf.d/mysqld.cnf -----□edit the file as,
(bind-address& 0.0.0.0
mysqlx-bind-address- 0.0.0.0)

```
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address            = 0.0.0.0
mysqlx-bind-address     = 0.0.0.0
#
# * Fine Tuning
```

After this do ctrl+x to save

Yes -y

enter

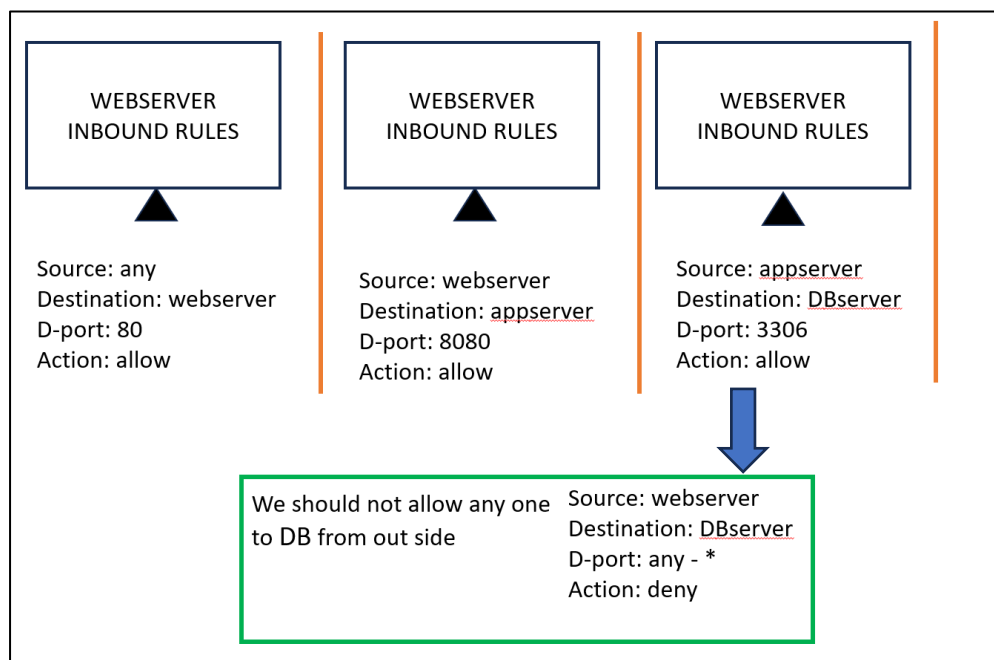
16. systemctl restart mysql -□restart my sql
17. service mysql restart --□ restart my sql service
18. systemctl status mysql.service --□check the status
19. after successful installation you will be able to see the putty console like below

```
[1]+ Stopped                               systemctl restart mysql
azadmin@DB-VM:~$ service mysql restart
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to restart 'mysql.service'.
Authenticating as: Ubuntu (azadmin)
Password:
==== AUTHENTICATION COMPLETE ====
^Z
[2]+ Stopped                               service mysql restart
azadmin@DB-VM:~$ systemctl status mysql.service
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2024-09-23 11:06:17 UTC; 19s ago
     Process: 4160 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 4170 (mysqld)
      Status: "Server is operational"
        Tasks: 38 (limit: 1064)
       Memory: 386.1M (peak: 403.9M)
          CPU: 931ms
         CGroup: /system.slice/mysql.service
                 └─4170 /usr/sbin/mysqld

Sep 23 11:06:14 DB-VM mysqld[4170]: mysqld: [ERROR] Found option without preceding group in config file /etc/mysql/mysql.conf.d/mysqld.cnf
Sep 23 11:06:14 DB-VM mysqld[4170]: mysqld: [ERROR] Stopped processing the 'includedir' directive in file /etc/mysql/my.cnf
Sep 23 11:06:14 DB-VM mysqld[4170]: 2024-09-23T11:06:14.937197Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.34) starting
Sep 23 11:06:14 DB-VM mysqld[4170]: 2024-09-23T11:06:14.945953Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started
Sep 23 11:06:16 DB-VM mysqld[4170]: 2024-09-23T11:06:16.656367Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended
Sep 23 11:06:17 DB-VM mysqld[4170]: 2024-09-23T11:06:17.473204Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed
```

now go to your azure portal write the NSG rules as below as per your requirement...

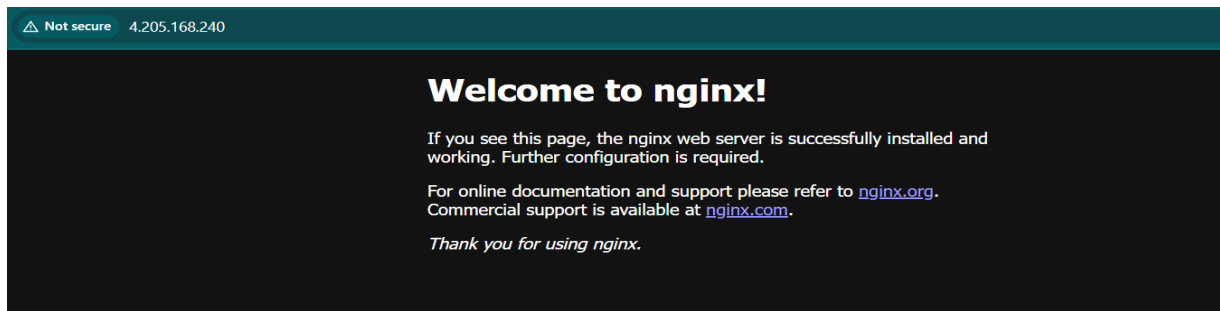
NSG RULES:



Results:

1. Anyone should be able to access WEB-VM on port 80 [it should be accessible via browser]

a) **Yes, connected to WEB-VM on 80 via browser.**



2. By using telnet command from WEB-VM, Result should be able to get connect to APP-VM on port 8080.

a) **Yes, connected to APP-VM on 8080 via WEB-VM.**

```
root@WEB-VM: /home/azadmin

Expanded Security Maintenance for Applications is not enabled.

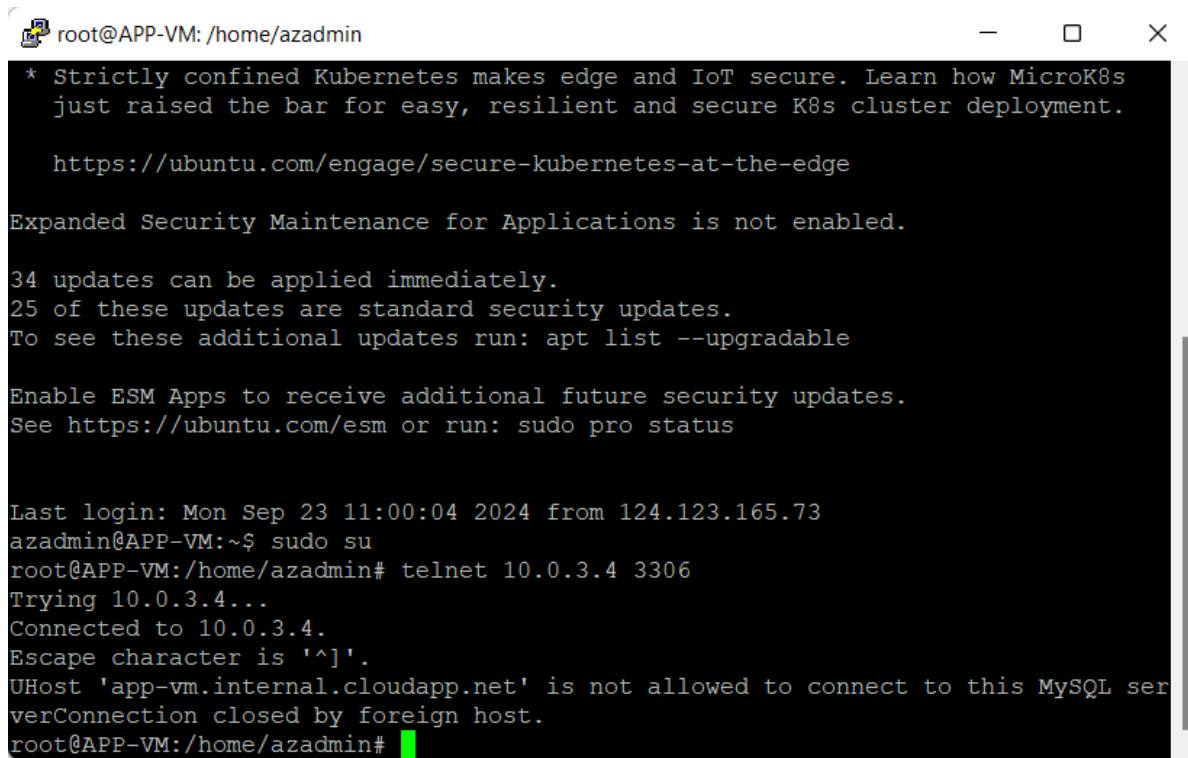
34 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep 23 10:43:51 2024 from 124.123.165.73
azadmin@WEB-VM:~$ sudo su
root@WEB-VM:/home/azadmin# telnet 10.0.2.4 8080
Trying 10.0.2.4...
Connected to 10.0.2.4.
Escape character is '^]'.
^ZConnection closed by foreign host.
root@WEB-VM:/home/azadmin# telnet 10.0.2.4 8080
Trying 10.0.2.4...
Connected to 10.0.2.4.
Escape character is '^]'.
^ZConnection closed by foreign host.
root@WEB-VM:/home/azadmin#
```

3. telnet from APP-VM to DB-VM server on port 3306/1433 result should be connected.

a. **Yes, connected to DB-VM on 3306 via APP-VM.**

A terminal window titled 'root@APP-VM: /home/azadmin' with standard window controls. The terminal output shows a security notice about MicroK8s, update information (34 updates available), and a successful telnet connection to 10.0.3.4 on port 3306. The connection message states 'Connected to 10.0.3.4.' and 'Escape character is '^]'.', followed by a MySQL error message: 'UHost 'app-vm.internal.cloudapp.net' is not allowed to connect to this MySQL serverConnection closed by foreign host.' The prompt returns to 'root@APP-VM: /home/azadmin#'.

```
root@APP-VM: /home/azadmin
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep 23 11:00:04 2024 from 124.123.165.73
azadmin@APP-VM:~$ sudo su
root@APP-VM:/home/azadmin# telnet 10.0.3.4 3306
Trying 10.0.3.4...
Connected to 10.0.3.4.
Escape character is '^]'.
UHost 'app-vm.internal.cloudapp.net' is not allowed to connect to this MySQL ser
verConnection closed by foreign host.
root@APP-VM:/home/azadmin#
```

4. By using telnet command from WEB-VM, Result should not get connect to DB-VM on any port
Result ⇒ no connection, or connection refused.

a. Yes, connection refused to DB-VM on 3306 via WEB-VM.

```
root@WEB-VM: /home/azadmin

34 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep 23 10:43:51 2024 from 124.123.165.73
azadmin@WEB-VM:~$ sudo su
root@WEB-VM:/home/azadmin# telnet 10.0.2.4 8080
Trying 10.0.2.4...
Connected to 10.0.2.4.
Escape character is '^]'.
^ZConnection closed by foreign host.
root@WEB-VM:/home/azadmin# telnet 10.0.2.4 8080
Trying 10.0.2.4...
Connected to 10.0.2.4.
Escape character is '^]'.
^ZConnection closed by foreign host.
root@WEB-VM:/home/azadmin# telnet 10.0.3.4 3306
Trying 10.0.3.4...
```